End Course Summative Assignment

Problem Statement: Write the Solutions to the Top 50 Interview Questions and Explain any 5 Questions in a Video

Video Link:

Questions

1. What is a vector in mathematics?

2. How is a vector different from a scalar?

3. What are the different operations that can be performed on vectors?

4. How can vectors be multiplied by a scalar?

5. What is the magnitude of a vector?

6. How can the direction of a vector be determined?

7. What is the difference between a square matrix and a rectangular matrix?

8. What is a basis in linear algebra?

9. What is a linear transformation in linear algebra?

10. What is an eigenvector in linear algebra?

11. What is the gradient in machine learning?

12. What is backpropagation in machine learning?

13. What is the concept of a derivative in calculus?

14. How are partial derivatives used in machine learning?

15. What is probability theory?

16. What are the primary components of probability theory?

17. What is conditional probability, and how is it calculated?

18. What is Bayes theorem, and how is it used?

19. What is a random variable, and how is it different from a regular variable?

20. What is the law of large numbers, and how does it relate to probability theory?

21. What is the central limit theorem, and how is it used?

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33. What is the relationship between the correlation coefficient and the covariance of a joint probability distribution?

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48. What are the key elements to consider when designing an experiment?

49. How can sample size determination affect experiment design?

50. What are some strategies to mitigate potential sources of bias in experiment design?

51. What is the geometric interpretation of the dot product?

52. What is the geometric interpretation of the cross-product?

53. How are optimization algorithms with calculus used in training deep learning models?

54. What are observational and experimental data in statistics?

55. How are confidence tests and hypothesis tests similar? How are they different?

56. What is the left-skewed distribution and the right-skewed distribution?

57. What is Bessel's correction?

58. What is kurtosis?

59. What is the probability of throwing two fair dice when the sum is 5 and 8?

60. What is the difference between Descriptive and Inferential Statistics?

61. Imagine that Jeremy took part in an examination. The test has a mean score of 160, and it has a standard deviation of 15. If Jeremy's z-score is 1.20, what would be his score on the test?

62. In an observation, there is a high correlation between the time a person sleeps and the amount of productive work he does. What can be inferred from this?

63. What is the meaning of degrees of freedom (DF) in statistics?

64. If there is a 30 percent probability that you will see a supercar in any 20-minute time interval, what is the probability that you see at least one supercar in the period of an hour (60 minutes)?

65. What is the empirical rule in Statistics?

66. What is the relationship between sample size and power in hypothesis testing?

67. Can you perform hypothesis testing with non-parametric methods?

68. What factors affect the width of a confidence interval?

69. How does increasing the confidence level affect the width of a confidence interval?

70. Can a confidence interval be used to make a definitive statement about a specific individual in the population?

71. How does sample size influence the width of a confidence interval?

72. What is the relationship between the margin of error and confidence interval?

73. Can two confidence intervals with different widths have the same confidence level?

74. What is a Sampling Error and how can it be reduced?

75. What is a Chi-Square test?

76. What is a t-test?

77. What is the ANOVA test?

78. How is hypothesis testing utilised in A/B testing for marketing campaigns?

79. What is the difference between one-tailed and two tailed t-tests?

80. What is an inlier?

Interview Questions and Answers

Q1. What is a Vector?

- A vector is a mathematical object that has both magnitude (size) and direction, represented as an ordered set of numbers like(x,y,z)
- Vectors are fundamental in ML for representing data points, weights, and feature embeddings in multi-dimensional space.

Q2. What is the difference between a vector and scalar?

- A scalar has only magnitude (temperature=30°C), while a vector has both magnitude and direction (velocity = 70km/h South).

- In machine learning , scalar represent single values like learning rate, while vectors represent feature sets or model weights.

Q3. What are vector operations?

- Key vector operations include addition, subtraction, scalar multiplication, dot product, and cross product.
- These operations are the building blocks of neural network computations, including forward pass and gradient calculations.

Q4. How can vectors be multiplied by a scalar?

- Scalar multiplication scales each component of the vector by a constant: k.(x,y) = (kx,ky).
- It changes the magnitude of the vector without altering its direction (unless k is negative, which flips it).

Q5. How do you calculate the magnitude of a vector?

- Magnitude is calculated as the square root of the sum of square components: |v| = {x2 + y2 + z2} .
- It represents the length or size of vector and is used to normalize vector into unit vectors

Q6. How do you find the direction of a vector?

- Direction is found by computing the unit vector: u = v / |v|, which has magnitude 1 and the same direction as v.
- Unit vectors are crucial in deep learning for normalizing inputs and computing cosine similarities.

Q7. What are different types of matrices?

- Square matrices have equal rows and columns; rectangular matrices do not. Both are used extensively in linear transformations.
- Special types include identity, diagonal, symmetric, and orthogonal
- matrices, each with unique mathematical properties.

Q8. What is a basis in linear algebra?

- A basis is a set of linearly independent vectors that span an entire vector space. Any vector in that space can be expressed as their combination.
- Choosing the right basis (PCA) helps reduce dimensionality while retaining the most important information.

Q9.What is a linear transformation?

- A linear transformation maps vectors to new vectors while preserving vector addition and scalar multiplication: $T(au + bv) = aT(u) + bT(v)$.
- It is represented as matrix multiplication, forming the foundation of how neutral network layers transform data.

Q10. What is an eigenvector?

- An eigenvector is a non zero vector whose direction remains unchanged after a linear transformation.
- Eigenvectors are central to PCA, where they define the principal components in data.

Q 11. (Original Q51) What is the geometric interpretation of the dot product?

- The dot product is $A.B = |A||B|\cos\theta$, producing a scalar that measures the degree of alignment between two vectors.

- It is used in attention mechanisms, cosine similarity, and computing projections in machine learning models.

Q12.(Original Q11) What is the gradient in machine learning?

- A gradient is a vector of partial derivatives that points in the direction of the steepest increase of a function.
- In model training, we move opposite to the gradient(gradient descent) to minimize the loss and update weights.

Q13. What is backpropagation in machine learning?

- Backpropagation is an algorithm that calculates the gradient of a neural network's error function.
- It enables efficient weight updates in neutral networks by telling each parameter how much it contributed to the error.

Q14. What is the concept of a derivative in calculus?

- A derivative measures the instantaneous rate of change of a function with respect to its input variable**.**
- In deep learning , derivatives tell us how much the loss changes when a weight is adjusted, guiding the optimization process.

Q15. What are partial derivatives and why are they used?

- A partial derivative measures the rate of change of a multi variable function with respect to one variable, holding all others constant.
- They are essential in gradient computation and used in backpropagation to determine how each weight affects the loss.

Q16. What is probability theory?

- Probability theory is the mathematical framework for quantifying uncertainty and measuring the likelihood of events occurring.

- It underpins all statistical inference, machine learning models and decision making under uncertainty.

Q17. What are the primary components of probability theory?

- A probability model consists of a sample space (all possible outcomes), events(subsets of outcomes), and probability rules (addition and multiplication).
- Together, these define how likely each event is, forming the foundation of both classical and Bayesian statistics.

Q18. What is conditional probability, and how is it calculated?

- Conditional probability is the likelihood of an event occurring given that another event has already occurred. It is the basis of Bayes theorem and widely used in spam filtering, medical diagnosis and recommendation systems.
- The mathematical formula is: $P(A|B) = P(A \cap B) / P(B)$

Q19. What is Bayes theorem, and how is it used?

- Bayes' Theorem is a mathematical formula used to update the probability of a hypothesis as more evidence becomes available. It allows you to reverse conditional probabilities, calculating the probability of a cause given an observed effect.
- The formula is written as: $P(A|B) = P(B|A). P(A) / P(B)$

Q20. What is a random variable, and how is it different from a regular variable?

- A random variable is a variable whose value depends on the outcome of a random phenomenon.

- Unlike a regular algebra variable that represents a fixed, hidden number, a random variable maps real-world uncertainties to numerical values before an event even happens.

Q21. What is the law of large numbers, and how does it relate to probability theory?

- The Law of Large Numbers (LLN) states that as the sample size increases, the sample mean converges to the true population mean.
- It justifies using sample statistics to estimate population parameters and is the theoretical basis for Monte Carlo simulations.

Q22. What is the central limit theorem, and how is it used?

- The Central Limit Theorem (CLT) states that the distribution of sample means approaches a normal distribution as sample size increases, regardless of the population's original shape.
- This is why the normal distribution appears everywhere in statistics. It describes the behavior of average people from any population.

Q23. What is the difference between discrete and continuous probability distributions?

- Discrete distributions (Binominal, Poisson) assign probabilities to countable outcomes
- Continuous distributions (Normal, Exponential) cover an infinite range of values.
- The key difference is that discrete uses probability mass function (PMF) while continuous uses probability density functions(PDF).

Q24. What are some common measures of central tendency, and how are they calculated?

- The three main measures of central tendency are the mean(average), median(middle value), and mode(most frequent) each summarizes data differently. They represent the center point of a dataset.

- Median is preferred when data is skewed or has outliers, as it is not influenced by extreme value the way the mean is.

Q25. What is the purpose of using percentiles and quartiles in data summarization?

- Percentiles and quartiles show the relative standing and distribution of data points within a dataset**.** While central tendency measures show the center, percentiles and quartiles help you understand the spread, rank, and presence of outliers.

Q26. How do you detect and treat outliers in a dataset?

- IQR Method: Calculate the Interquartile Range IQR = Q3 - Q1. Any data point below
- Q1 - 1.5 * *IQR or above Q3 + 1.5* * IQR is classified as an outlier.
- Z-Score Method: Measure how many standard deviations a data point is from the mean. Values with a Z-score below -3 or above +3are considered outliers.

Q27. How do you use the central limit theorem to approximate a discrete probability distribution?

- To approximate a discrete distribution using the Central Limit Theorem (CLT), you calculate the discrete distribution's mean (μ) and standard deviation (σ), then fit them to a continuous normal curve**.**

Q28. How do you test the goodness of fit of a discrete probability distribution?

- You test the goodness of fit of a discrete probability distribution using the Chi-Square Goodness-of-Fit Test**.**

Q29. What is a joint probability distribution?

- A joint probability distribution is a statistical function that calculates the likelihood of two or more random variables occurring simultaneously.

Q30. How do you calculate the joint probability distribution?

- Joint probability is computed as P(A, B) = P(A|B) * P(B), derived directly from the definition of conditional probability.
- For independent events, this simplifies to P(A, B) = P(A) * P(B), since knowing B gives no information about A.

Q31. What is the difference between a joint probability distribution and a marginal probability distribution?

- Joint probability captures the combined probability of multiple events, marginal probability of a single event ignoring all others.
- Marginal probability is obtained by summing (or integrating) the joint probability over all values of the other variable.

Q32. What is the covariance of a joint probability distribution?

- The covariance of a joint probability distribution is a metric that measures the directional linear relationship between two random variables. It indicates whether the variables tend to grow together or move in opposite directions.

Q33. How do you determine if two random variables are independent based on their joint probability distribution?

- To determine if two random variables are independent, you check if their joint probability is equal to the product of their individual marginal probabilities for every single possible outcome.

Q34. What is the relationship between the correlation coefficient and the covariance of a joint probability distribution?

- The correlation coefficient is a normalized version of covariance. While covariance tells you the direction of the relationship between two variables, the correlation coefficient scales that number down to a strict range between -1 and +1 to tell you the actual strength.

Q35. What is sampling in statistics, and why is it important?

- Sampling is the process of selecting a smaller, representative subset of individuals from a larger population to estimate the characteristics of the

whole group. It allows researchers to uncover reliable population truths without needing to analyze every single entity.

Q36. What are the different sampling methods commonly used in statistical inference?

- Sampling methods are divided into two main categories: probability sampling (every member has a known, non-zero chance of selection) and non-probability sampling (selection relies on subjective judgment).

Q37. What is the central limit theorem, and why is it important in statistical inference?

- The Central Limit Theorem (CLT) states that if you take large, random samples from any population, the distribution of the sample means will follow a normal distribution (a bell curve) [CLT]. This holds true regardless of the original population's shape, whether it is skewed, uniform, or completely random [CLT].

Q38. What is the difference between parameter estimation and hypothesis testing?

- The primary difference lies in the ultimate goal of the statistical analysis:
- Parameter Estimation: Focuses on measuring an unknown value (like calculating the exact average income of a city based on sample data).
- Hypothesis Testing: Focuses on making a yes-or-no decision (like determining if a new marketing campaign significantly increased sales compared to the old one).

Q39. What is the p-value in hypothesis testing?

- A p-value is the probability of obtaining test results at least as extreme as the observed data, assuming that the null hypothesis (the status quo) is completely true.

Q40. What is confidence interval estimation?

- A confidence interval estimation is a range of values, calculated from sample data, that is likely to contain the true population parameter.

Q41. What are Type I and Type II errors in hypothesis testing?

Type I and Type II errors are the two specific ways a statistical decision can be wrong when running a hypothesis test:

- Type I Error (False Positive): Rejecting the null hypothesis when it is actually true. You conclude an effect or difference exists when it does not (e.g., convicting an innocent person).
- Type II Error (False Negative): Failing to reject the null hypothesis when it is actually false. You miss an effect or difference that actually exists (e.g., acquitting a guilty person).

Q42. What is the difference between correlation and causation?

The core difference is that correlation measures a relationship, while causation proves a cause-and-effect link:

- Correlation: Two variables move or change together over time (e.g., as ice cream sales rise, shark attacks also rise).
- Causation: A change in one variable directly forces a change in the other variable (e.g., turning a heater on causes the room temperature to rise).

Q43. How is a confidence interval defined in statistics?

- A confidence interval is a range of values derived from sample data that is likely to contain the true, unknown population parameter. Rather than offering a single estimate, it establishes a mathematically calculated buffer zone that accounts for sampling variability.

Q44. What does the confidence level represent in a confidence interval?

- The confidence level represents the long-term success rate of the estimation method across repeated sampling**.** Specifically, a 95% confidence level means that if you take 100 independent random samples from the same population and calculate an interval for each, approximately 95 of those intervals will successfully contain the true population parameter.

Q45. What is hypothesis testing in statistics?

- Hypothesis testing is a formal statistical process used to decide whether a theory about a population is supported by sample evidence.

Q46. What is the purpose of a null hypothesis in hypothesis testing?

- The purpose of a null hypothesis is to serve as a skeptical baseline assumption that there is no effect, no change, or no relationship in the population.

Q47. What is the difference between a one-tailed and a two-tailed test?

- The core difference lies in the directionality of the alternative hypothesis and where the rejection region is located on the probability curve:

Q48. What is experiment design, and why is it important?

- Experimental design is the process of planning a research study to ensure that the data collected can answer a specific question clearly and efficiently.

Q49. What are the key elements to consider when designing an experiment?

- When designing an experiment, you must clearly isolate your independent and dependent variables while keeping all other outside factors strictly controlled. You then apply core design principles like randomization, replication, and control groups to eliminate human bias. This systematic approach ensures your final data yields a valid, cause-and-effect conclusion rather than a random fluke.

Q50. What are some strategies to mitigate potential sources of bias in experiment design?

- To mitigate bias, experiments must use randomized assignment to distribute hidden variables evenly and blinding protocols to prevent human expectations from skewing data. Additionally, maintaining identical testing conditions eliminates outside procedural variance, while pre-registering the study prevents data cherry-picking.